

NVIDIA Corporation

Sustained Spending Mix Shift Towards GPU On Proliferation of AI/Accelerated Compute Applications; Driving Higher Software Monetization In AI

NVIDIA held its financial analyst event yesterday following the GTC keynote presentation (see our takeaways [here](#)) from Monday. The team highlighted its strong ecosystem (silicon, system hardware, software, customer partnerships, installed user/developer base) to address the growing opportunities in its data center end market. AI/accelerated computing build-out continues to drive spending mix shift towards GPU/accelerators and team believes that it can continue to capture a higher % of the \$250B in annual datacenter infrastructure spending (going to \$500B LT) vs. historical given the shift to accelerated computing, strong portfolio breadth(GPU/CPU/Networking) and system-solution approach (higher \$\$ content vs. chips only). The team believes that GPU is the right architecture given that it is optimized for a broad spectrum of AI workloads (vs. custom ASICs). Demand for GPU remains strong and the team sees a strong runaway for greenfield investment before we get into a upgrade/replacement cycle. Beyond the hardware, Nvidia remains focused on its software strategy which helps it open up new markets. The team is also driving higher software monetization in the AI layer with NIM for enterprise customers. NVIDIA Inference Microservice (NIM) are pre-trained models (e.g. computer vision, speech recognition, robotics) that are packaged and optimized to run across its installed base for its enterprise customers. Cost of running are \$4500 per GPU per year and the team is planning to deploy hundreds of different models, which we expect will driven strong revenue growth for its software business. The team is also working with companies to create AI tools such as copilots. In the near term, new products are expected to come to market later this year. The team has been working diligently with its supply chain partners to prepare for this volume ramp. Given some of the demand signals from its customers, the team expects to be supply constrained for its new products. We are encouraged by the market growth opportunities and, more importantly, NVIDIA's ability to leverage its silicon architecture across multiple end-markets and applications, supported by the company's investments in software and ecosystem development, to drive GM/OPM/EPS expansion. We reiterate our Overweight rating on NVDA

- **Capturing a higher % of datacenter infrastructure spending.** The team believes that it can continue to capture a higher % of the \$250B in annual datacenter infrastructure spending (going to \$500B LT) vs. historical given the shift to accelerated computing, strong portfolio breadth(GPU/CPU/Networking) and system-solution approach (higher \$\$ content vs. chips only). The team sees a strong runaway for greenfield investments before we get into a upgrade/replacement cycle.

Overweight

NVDA, NVDA US
Price (19 Mar 24):\$893.98
Price Target (Dec-24):\$850.00

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Quarterly Forecasts (FYE Jan)

Adj. EPS (\$)	2024A	2025E	2026E
Q1	1.09	5.32	
Q2	2.70	5.49	
Q3	4.02	5.80	
Q4	5.16	6.09	
FY	12.96	22.70	

Style Exposure

Quant Factors	Current %Rank	Hist %Rank (1=Top)			
		6M	1Y	3Y	5Y
Value	81	88	84	77	79
Growth	25	80	71	74	33
Momentum	5	1	46	55	93
Quality	2	7	6	7	2
Low Vol	58	63	63	45	72
ESGQ	21	9	11	2	1

Sources for: Style Exposure – J.P. Morgan Quantitative and Derivatives Strategy; all other tables are company data and J.P. Morgan estimates.

See page 6 for analyst certification and important disclosures.

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Price Performance



	YTD	1m	3m	12m
Abs	78.6%	21.8%	76.6%	243.8%
Rel	70.6%	18.9%	68.0%	213.5%

Company Data

Shares O/S (mn)	2,499
52-week range (\$)	974.00-251.30
Market cap (\$ mn)	2,234,056.00
Exchange rate	1.00
Free float(%)	96.1%
3M - Avg daily vol (mn)	50.92
3M - Avg daily val (\$ mn)	35,212.9
Volatility (90 Day)	44
Index	S&P 500
BBG BUY HOLD SELL	60 7 0

Key Metrics (FYE Jan)

\$ in millions	FY24A	FY25E
Financial Estimates		
Revenue	60,922	104,142
Adj. EBIT	33,069	63,685
Adj. EBITDA	33,493	64,109
Adj. net income	32,312	56,606
Adj. EPS	12.96	22.70
BBG EPS	24.65	29.31
Cashflow from operations	28,091	48,207
FCFF	27,023	46,907
Margins and Growth		
Revenue Growth Y/Y (%)	125.9%	70.9%
EBIT margin	54.3%	61.2%
EBIT growth	464.5%	92.6%
EBITDA margin	55.0%	61.6%
EBITDA Growth Y/Y (%)	433.2%	91.4%
Net margin	53.0%	54.4%
Adj. EPS growth	288.4%	75.2%
Ratios		
Adj. tax rate	12.1%	17.0%
Interest cover	-	-
Net debt/Equity	NM	NM
Net debt/EBITDA	NM	NM
ROE	99.3%	80.0%
Valuation		
FCFF yield	1.2%	2.1%
Dividend yield	0.1%	0.1%
EV/Revenue	27.4	15.6
EV/EBITDA	49.9	25.3
Adj. P/E	69.0	39.4

Summary Investment Thesis and Valuation

Investment Thesis

We believe NVIDIA continues to execute across all segments. While 1H is typically seasonally weaker than 2H, we expect solid demand in PC gaming to be a strong revenue driver for the company, offsetting PC OEM, which is in secular decline. We expect the data center segment to grow strongly as hyperscale customers continue to embrace GPU-accelerated deep learning for processing large data sets. We are encouraged by strength in the automotive and enterprise segments as well, although strong adoption of autonomous driving in the market remains to be seen. We anticipate significant upside in the shares, driving our Overweight rating.

Valuation

Our Dec-24 PT assumes that NVIDIA trades at 30-35x applied to ~\$29.00 of earnings power in CY25, discounted back to CY24 assuming a 12% discount rate. Our outlook of \$29.00 of earnings power assumes the team will drive its earnings power by 30%+ per year over the next few years led by the continued strong datacenter growth (30-40% CAGR), monetizing an incremental ~\$14B of auto revenue pipeline and an incremental \$2-3B of software, licensing, and subscription revenues over the next 3-4 years.

Performance Drivers



Factors	6M Corr	1Y Corr
Market: MSCI US	0.72	0.53
Sect: Technology	0.62	0.76
Ind: Semicond & S Equip	0.78	0.88
Macro:		
Economic Surprise	0.23	0.26
Crude Oil	-0.24	-0.24
US 10yr Breakeven	-0.13	-0.20
Quant Styles:		
DivYld	-0.62	-0.42
Value	-0.39	-0.32
Growth	0.24	0.27

Source: J.P. Morgan Quantitative and Derivatives Strategy for Performance Drivers; company data, Bloomberg Finance L.P. and J.P. Morgan estimates for all other tables. Note: Price history may not be complete or exact.

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- **Well-positioned to support strong product ramp later this year; expect supply constraints on strong demand uptake.** New products are expected to come to market later this year (H200/B100/B200). The team has been working diligently with its supply chain partners to prepare for this volume ramp. Given some of the demand signals from its customers, the team expects to be supply constrained for its new products.
- **Drive higher software monetization in the AI layer.** Nvidia remains a software led company which helps the team open up new markets. The team is also driving higher software monetization in the AI layer with NIM for enterprise customers. NVIDIA Inference Microservice (NIM) are pre-trained models (e.g. computer vision, speech recognition, robotics) that are packaged and optimized to run across its installed base for its enterprise customers. Cost of running are \$4500 per GPU per year and the team is planning to deploy hundreds of different models which we expect will driven strong growth in its software business.

Investment Thesis, Valuation and Risks

NVIDIA Corporation (*Overweight; Price Target: \$850.00*)

Investment Thesis

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Valuation

Our Dec-24 PT assumes that NVIDIA trades at 30-35x applied to ~\$29.00 of earnings power in CY25E, discounted back to CY24 assuming a 12% discount rate. Our outlook of \$29.00 of earnings power assumes the team will drive its earnings power by 30%+ per year over the next few years led by the continued strong datacenter growth (30-40% CAGR), monetizing an incremental ~\$14B of auto revenue pipeline and an incremental \$1-2B of software, licensing, and subscription revenues over the next 3-4 years.

Risks to Rating and Price Target

- Although PC gaming demand seems resilient to macroeconomic weakness, any macro uncertainty could weigh on PC gaming demand trends. Given NVDA's ~53% exposure to the PC gaming segment, any consumer PC gaming weakness poses downside risk to our estimates.
- NVIDIA's GPUs could gain lower than expected deployment into data center applications as hyperscale customers further adopt deep learning as a new and effective way of processing large unstructured data sets. Any significant decrease in the adoption of deep learning by hyperscale customers or increase in competition could result in downside risk to our revenue and earnings estimates.

NVIDIA Corporation: Summary of Financials

Income Statement - Annual						Income Statement - Quarterly					
	FY23A	FY24A	FY25E	FY26E	FY27E		1Q25E	2Q25E	3Q25E	4Q25E	
Revenue	26,974	60,922	104,142	-	-	Revenue	24,000	25,461	26,693	27,988	
COGS	(11,618)	(16,621)	(26,093)	-	-	COGS	(5,688)	(6,543)	(6,780)	(7,081)	
Gross profit	15,356	44,301	78,050	-	-	Gross profit	18,312	18,917	19,913	20,907	
SG&A	(2,160)	(2,558)	(3,036)	-	-	SG&A	(731)	(746)	(761)	(799)	
Adj. EBITDA	6,282	33,493	64,109	-	-	Adj. EBITDA	14,939	15,474	16,399	17,297	
D&A	(424)	(424)	(424)	-	-	D&A	(106)	(106)	(106)	(106)	
Adj. EBIT	5,858	33,069	63,685	-	-	Adj. EBIT	14,833	15,368	16,293	17,191	
Net Interest	-	-	-	-	-	Net Interest	-	-	-	-	
Adj. PBT	5,864	33,680	64,685	-	-	Adj. PBT	15,083	15,618	16,543	17,441	
Tax	186	(4,059)	(10,997)	-	-	Tax	(2,564)	(2,655)	(2,812)	(2,965)	
Minority Interest	-	-	-	-	-	Minority Interest	-	-	-	-	
Adj. Net Income	8,365	32,312	56,606	-	-	Adj. Net Income	13,248	13,692	14,460	15,206	
Reported EPS	1.74	11.94	21.34	-	-	Reported EPS	4.98	5.15	5.46	5.75	
Adj. EPS	3.34	12.96	22.70	-	-	Adj. EPS	5.32	5.49	5.80	6.09	
DPS	0.68	0.68	0.68	-	-	DPS	0.17	0.17	0.17	0.17	
Payout ratio	39.0%	5.7%	3.2%	-	-	Payout ratio	3.4%	3.3%	3.1%	3.0%	
Shares outstanding	2,507	2,493	2,493	-	-	Shares outstanding	2,491	2,492	2,494	2,496	
Balance Sheet & Cash Flow Statement						Ratio Analysis					
	FY23A	FY24A	FY25E	FY26E	FY27E		FY23A	FY24A	FY25E	FY26E	FY27E
Cash and cash equivalents	13,296	25,984	72,495	-	-	Gross margin	56.9%	72.7%	74.9%	-	-
Accounts receivable	3,827	9,999	14,598	-	-	EBITDA margin	23.3%	55.0%	61.6%	-	-
Inventories	5,159	5,282	9,945	-	-	EBIT margin	21.7%	54.3%	61.2%	-	-
Other current assets	791	3,080	3,080	-	-	Net profit margin	31.0%	53.0%	54.4%	-	-
Current assets	23,073	44,345	100,118	-	-	ROE	34.3%	99.3%	80.0%	-	-
PP&E	3,807	3,914	4,790	-	-	ROA	19.6%	60.4%	60.2%	-	-
LT investments	-	-	-	-	-	ROCE	16.1%	67.8%	65.7%	-	-
Other non current assets	14,302	17,469	17,469	-	-	SG&A/Sales	8.0%	4.2%	2.9%	-	-
Total assets	41,182	65,728	122,377	-	-	Net debt/equity	NM	NM	NM	-	-
Short term borrowings	1,250	1,250	1,250	-	-	P/E (x)	268.0	69.0	39.4	-	-
Payables	1,193	2,699	3,807	-	-	P/BV (x)	101.4	51.9	22.6	-	-
Other short term liabilities	4,120	6,682	6,682	-	-	EV/EBITDA (x)	268.0	49.9	25.3	-	-
Current liabilities	6,563	10,631	11,739	-	-	Dividend Yield	0.1%	0.1%	0.1%	-	-
Long-term debt	9,703	8,459	8,459	-	-	Sales/Assets (x)	0.6	1.1	1.1	-	-
Other long term liabilities	2,815	3,660	3,660	-	-	Interest cover (x)	-	-	-	-	-
Total liabilities	19,081	22,750	23,858	-	-	Operating leverage	(19426.5%)	369.1%	130.5%	-	-
Shareholders' equity	22,101	42,978	98,520	-	-	Revenue y/y Growth	0.2%	125.9%	70.9%	-	-
Minority interests	-	-	-	-	-	EBITDA y/y Growth	(41.6%)	433.2%	91.4%	-	-
Total liabilities & equity	41,182	65,728	122,377	-	-	Tax rate	3.2%	12.1%	17.0%	-	-
BVPS	8.81	17.24	39.51	-	-	Adj. Net Income y/y Growth	(25.7%)	286.3%	75.2%	-	-
y/y Growth	(16.0%)	95.6%	129.2%	-	-	EPS y/y Growth	(24.9%)	288.4%	75.2%	-	-
Net debt/(cash)	(2,343)	(16,275)	(62,786)	-	-	DPS y/y Growth	0.0%	0.0%	0.0%	-	-
Cash flow from operating activities	5,642	28,091	48,207	-	-						
o/w Depreciation & amortization	424	424	424	-	-						
o/w Changes in working capital	(3,174)	(5,057)	(10,370)	-	-						
Cash flow from investing activities	7,374	(10,566)	(1,300)	-	-						
o/w Capital expenditure	(1,833)	(1,068)	(1,300)	-	-						
as % of sales	6.8%	1.8%	1.2%	-	-						
Cash flow from financing activities	(11,617)	(13,634)	(396)	-	-						
o/w Dividends paid	(398)	(396)	(396)	-	-						
o/w Net debt issued/(repaid)	(7)	(1,279)	0	-	-						
Net change in cash	1,399	3,891	46,511	-	-						
Adj. Free cash flow to firm	3,809	27,023	46,907	-	-						
y/y Growth	(53.2%)	609.5%	73.6%	-	-						

Source: Company reports and J.P. Morgan estimates.

Note: \$ in millions (except per-share data). Fiscal year ends Jan. o/w - out of which

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NVIDIA Corporation (NVDA, NVDA US) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Apr 04, 2002. All share prices are as of market close on the previous business day.

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